From "Where is Changshan?" to "How is Chinatown Seen?": Reflections on GIS and Spatial History

I first encountered GIS in middle school geography class. Back then, I saw it as a surface survey technology just like GPS and BeiDou, and thought it had little connection to history. This week’ s readings made me realize that digital tools not only expand the range of source materials, but also shape who can provide evidence and what experiences can be seen. This reminds me of what we discussed in HY6101: *interviewees are co-producers of knowledge*. Yet co-production does not equal equality. Issues of access to equipment, platform moderation, and interpretive authority still need to be examined.

While compiling historical materials on the medicinal herb *Changshan*, I gradually developed a geographical understanding: the "place of origin" is not necessarily a single coordinate on a map. Herb names and aliases, record keepers, circulation routes, and processing and usage contexts can all place the same object in different spatial relations. A dot on the map is only a clue. I also need to ask who named this place, on what basis, and what spatial experiences were left unrecorded. GIS truly serves historical inquiry only when it preserves these layers and ambiguities.

The readings offer me a method: translating concepts into data for verification. Warf and Sui remind me that maps are not neutral mirrors of the world, and crowdsourcing does not automatically equal democratization. White sees visualization as a research tool for raising questions and iterative revision. The *where*, *what*, metadata, precision and accuracy mentioned in bookdown are starting points for operation, but they do not answer the question of who defines "where." I no longer treat the correspondence between maps and facts as the only standard of truth. Instead, I aim for traceable sources and negotiable differences.

This shift can be applied to my ongoing "Urban Memory" project. We take Chinatown as our field site. On weekends, we conduct street observations and merchant interviews, and plan to design a 45 – 60 minute bilingual cultural route with five stops. Our preliminary observations suggest gaps between different sections of the neighborhood, and between tourism and daily commerce. We also have not found an official unified tourist map for the area. These are all observations yet to be verified. GIS should not only serve as a base map for navigation, but act as a research interface for comparing different memories.

If time permits later, I would like to further advance this project with the knowledge I have gained. I plan to create separate layers for historic buildings, religious spaces, merchants and food, oral memories, walking routes, and current operational status. Time fields will be used to show relocations, renovations, and changes in use. For each point, I will record original name, site type, time period, source, authorization status, confidence level, geometric/positional scale, and geocoding method. If oral accounts can only be

located to a street segment, they will not be forced into precise coordinates. Official records, oral materials, and tourist feedback will be kept separate, rather than merged into a single "popularity" value. Such a dataset is suitable for comparing place narratives and spatial changes, but cannot represent all tourists or residents.

According to an interviewee at the Baixinchun tea shop, the store has relocated four times due to government projects, but has always remained within Chinatown. Tea leaves are imported from mainland China, blended locally, and sold under the shop's own brand. This is not a static shop location, but a time-space relation woven together by policy, transportation, commodities, and family memory. Before public use, authorization must be obtained, addresses verified, and sources cross-checked. This also helps me refine the question from my Changshan study — *who defines the place of origin* — into *who defines Chinatown, and through what scale*.

Different tools answer different questions. GIS shows which memories are tied to places; timelines show when changes occurred; walkability networks compare which areas are bypassed; interviews and close reading of documents explain why the same place carries different meanings. QR codes can link points to bilingual story cards and audio. Static maps, legends, filters, event tables, alternative text, data dictionaries, and version records allow readers to trace from maps back to evidence. A printable version will also be provided for times with poor internet access. Route choice and stay duration data only represent voluntary participants, and cannot be generalized to all tourists. Gaps in data should also be marked.

Digitization also reshapes the distribution of narrative authority. Official attraction categorizations, rankings on travel platforms, and modern road boundaries may overshadow old merchant names, immigrant family memories, or religious norms. QR codes can also turn personal experiences into consumable labels. Therefore, the purpose of research should be stated before interviews, and separate consent obtained for recording, public citation, and location data. Participants should be allowed anonymity, revision, and withdrawal. Unauthorized content will only be used for internal analysis. When involving religious rituals, photographs, and music, relevant guidelines must be followed and copyright checked. Community members should be involved in naming, verification, and review, while managers only review aggregated trends.

Both this week's readings and my field practice have led me to reflect that in digital history and new geography, scholars care not just about *where a place is*, but about how places are continuously produced by objects, human actions, institutions, and narratives. Digital history is not about letting software deliver conclusions for me. It is about making relations and uncertainties open to questioning, verification, and collaborative revision.

Statement: This work was originally written in Chinese and subsequently translated using the AI tool ChatGPT.